

SYNTHETIC LETHALITY OF RPP25 AND RPP25L IN GLIOBLASTOMA USING CRISPR-CAS9 KNOCK OUT

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Abstract

This study explores the synthetic lethality of RPP25 and RPP25L in glioblastoma. The summer internship focused on three objectives: First, assessed the effects of knocking out RPP25 and RPP25L in different cancer cells. Second, evaluated the double knockout's efficiency to determine the synthetic lethality. Finally, investigated the rescue effect of RPP25L in GBM cell lines using the GFP competition assay, infected with RPP25 overexpression cells.

Introduction

Research shows that glioblastoma cell lines, such as U251 and LN229, have low RPP25 expression, while other cancer lines like HGC27 and A549 maintain adequate levels. RPP25 and RPP25L show genetic redundancy. This study aims to knock out RPP25L to assess its impact on glioblastoma proliferation. Given the low RPP25 expression in glioblastoma, CRISPR-Cas9-mediated knockout of RPP25L may induce synthetic lethality, leading to impaired proliferation in these cancer cells. In contrast, normal cells can compensate for RPP25L loss due to preserved RPP25 function, leaving their activity unaffected. This strategy offers a promising avenue for glioblastoma treatment.

Results

Figure 4. The overall knockout efficiency of RPP25Lsg1 is higher than that of RPP25Lsg2, while RPP25sg2 exhibits a higher knockout efficiency than RPP25sg1. Thus, select RPP25Lsg1 and RPP25sg2 for subsequent infection and GFP competition.

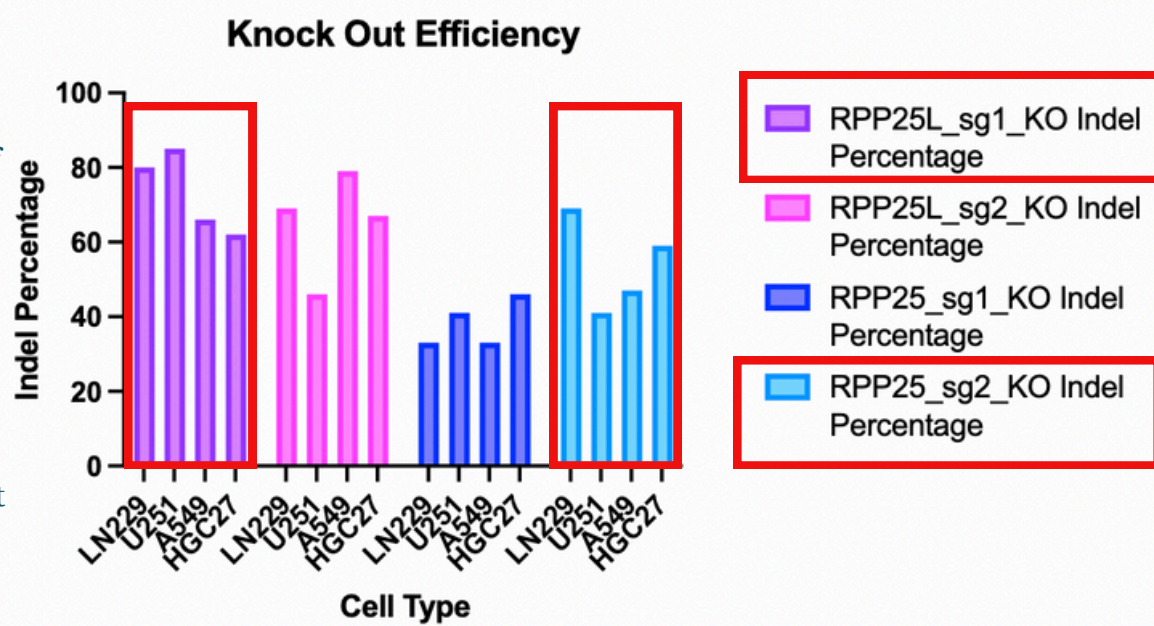


Figure 5. Mix 70% dual RNA cells with 30% wild-type cells, and track the relative abundance of dual RNA cells (GFP/BFP double positive) over time (day 0, day 7, and day 14) using FACS for GFP analysis.

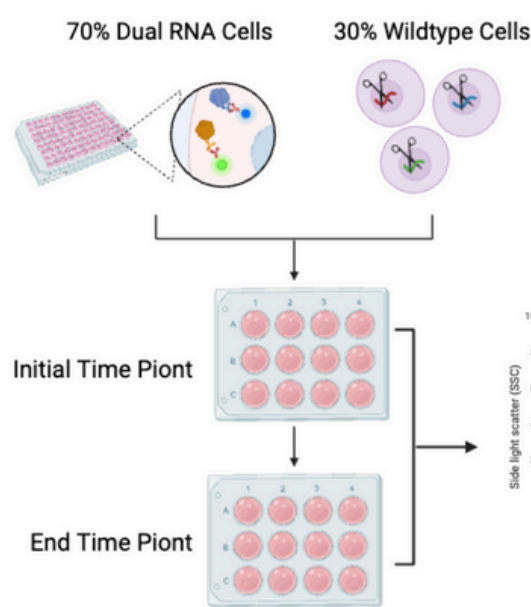
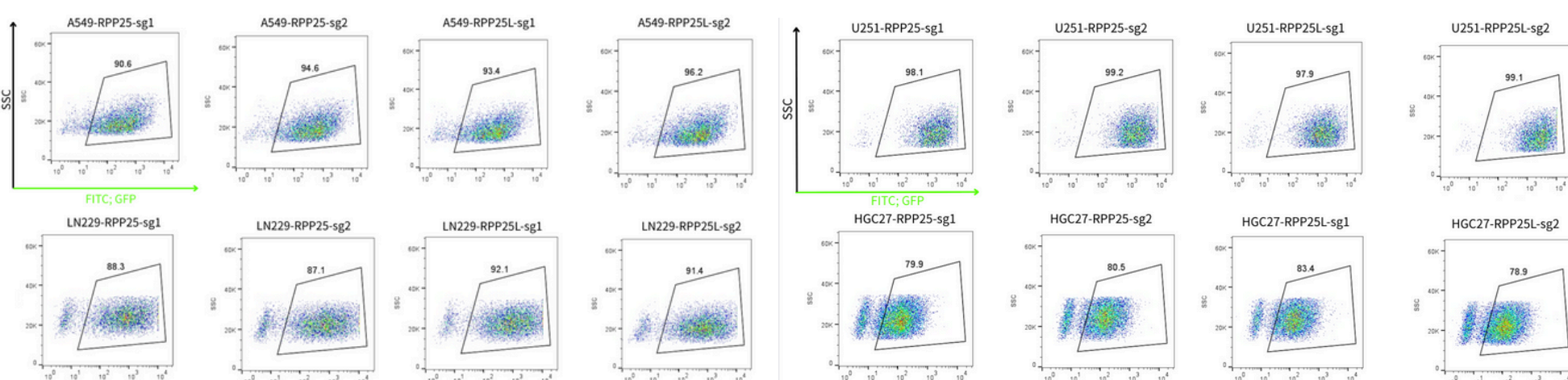


Figure 8. * $P < 0.05$; ** $P < 0.005$; *** $P < 0.0005$; ns(no significance)

$$\text{relative percentage} = \frac{++ve \text{ percentage of cell line condition end time point}}{++ve \text{ percentage of cell line condition initial time point}}$$
$$++ve \text{ percentage normalized to dummy} = \frac{\text{relative percentage of cell line condition}}{\text{relative percentage of cell line dummy}}$$

Figure 7. Equation of double positive percentage normalized to dummy.



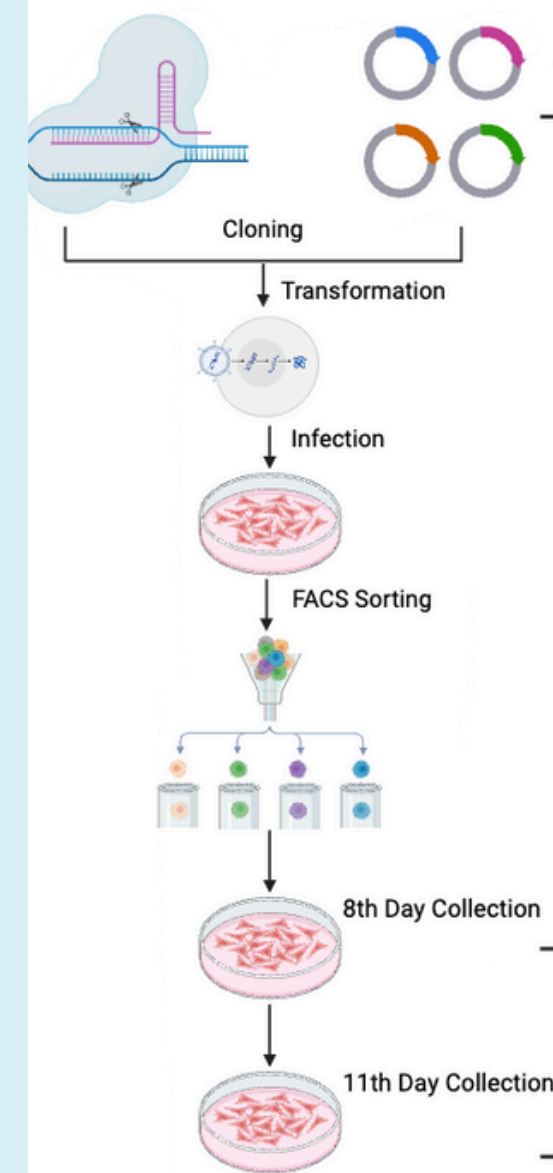
Conclusion

RPP25 and RPP25L show gene redundancy and synthetic lethality. And RPP25L is a promising novel therapeutic target in glioblastomas with hypermethylation of the RPP25 promoter. In the future, selective therapeutic inhibition of the remaining gene is feasible because non-tumor cells still retain the genetic buffer to tolerate the inhibition without systemic side effects. Additionally, RPP25 can be tractable biomarker in GBM, as measuring loss of paralog RPP25L in tumors permits selection of patients that would respond to inhibition of the synthetic lethal paralog RPP25.

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Methodology



Next-generation Sequencing
KO Efficiency Analysis

Figure 2. Experimental workflow for genome-wide CRISPR screen in RPP25 and RPP25L gene in LN229, A549, U251, and HGC27.

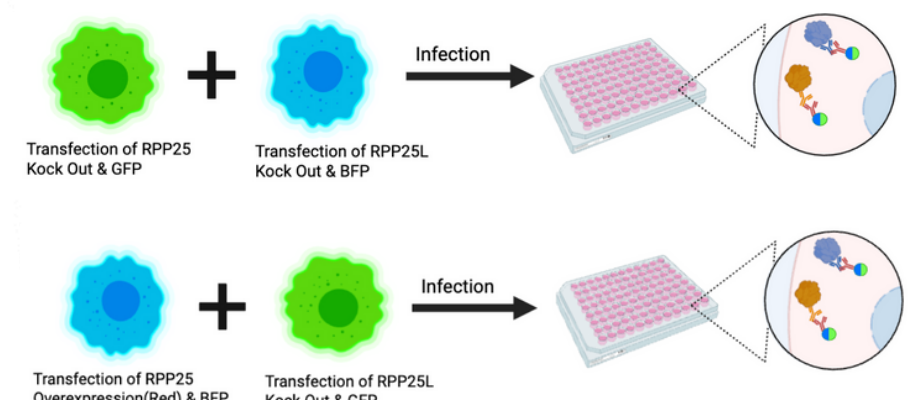


Figure 3. Visualization of synthetic lethality of RPP25 & RPP25L by using GFP competition assay. RPP25 KO cells and RPP25 OE cells were infected with RPP25L KO cells. Cell fusion was visualized by GFP signal.

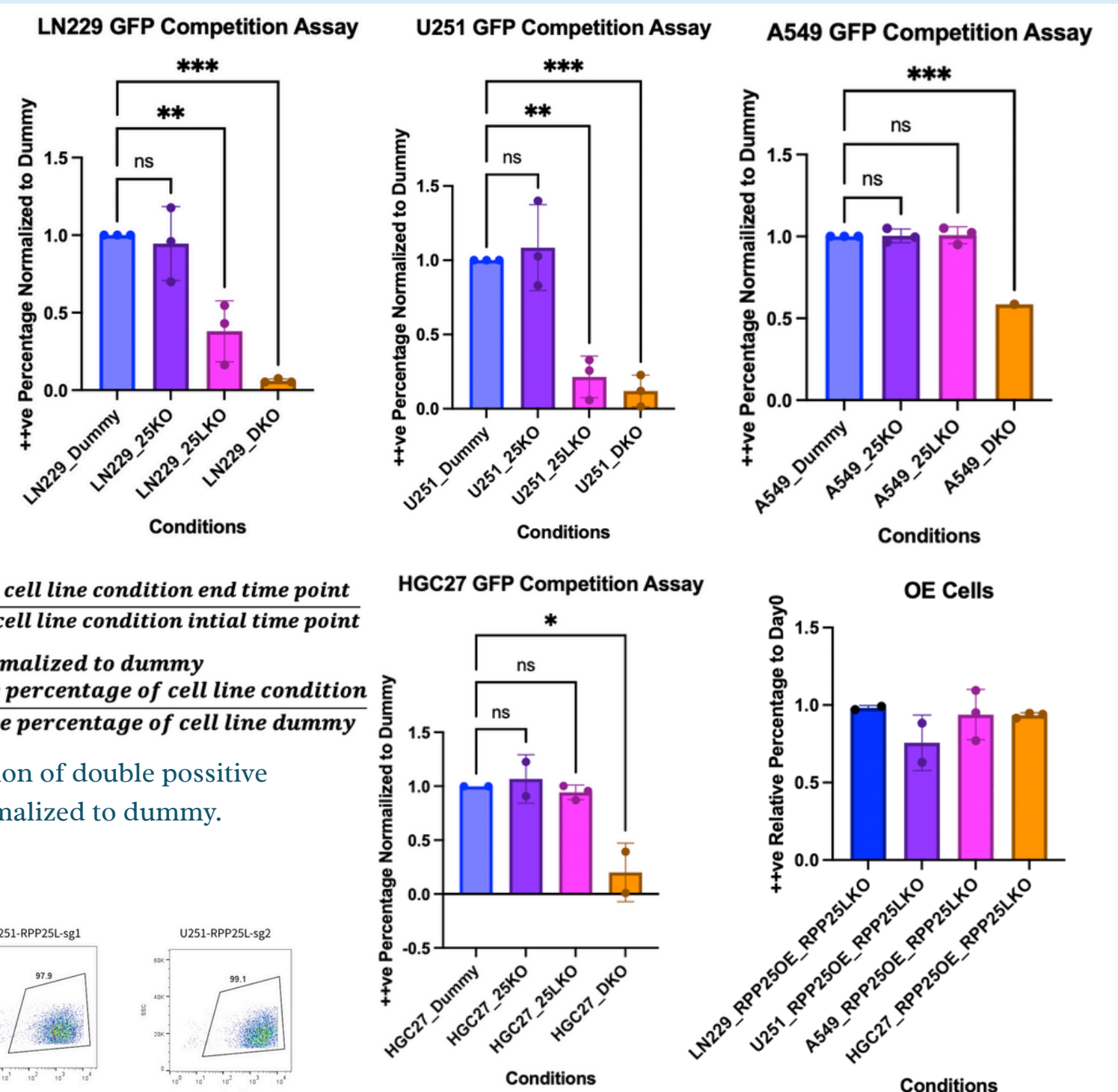


Figure 8. As RPP25LKO does not affect the control group, which means while RPP25L is knocked out, RPP25 can compensate and does not affect the phenotype of the cells. The overexpression of RPP25 can fully rescue the KO of RPP25L.

Figure 6. The figure shows the FACS results of edited cells, ranging from 78.9 to 99.2. Y-axis is SSC, X-axis is FITC-H, GFP. Edited cells marked by GFP, which are relatively high and pure.

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